

Lab 4 Assignment: Use LLM API for Meta Analysis

Course Name: CSS5120: Computational Linguistics Deadline (DDL): April 3rd 23:59
Total Score: 10 points

1. Background & Objectives

This course is about to enter the Final Project phase. To conduct high-quality research, students need to comprehensively review and understand the frontier literature related to their chosen topics. Faced with a massive volume of academic papers, traditional reading and information extraction can be highly time-consuming and labor-intensive.

Therefore, this experiment aims to introduce and allow you to practice how to utilize Large Language Models (e.g., GPT-4o-mini) as automated research assistants for efficient literature information integration and approximate Meta-analysis. You will write and modify a Python script, call the LLM API to automatically extract key data from multiple PDF papers, and structure it into a data table.

Given the high interest among students in LLM-related research (e.g., evaluating AI behaviors such as Theory of Mind, empathy, and role-playing), we have provided a sample table and extraction fields based on this theme in the guide. However, this is merely an example! We highly encourage students to fine-tune and customize the final extracted information fields (Output Schema) according to the actual research direction and needs of your Final Project.

Through this lab, you will master: * Basic Prompt Engineering techniques. * Methods for batch processing unstructured documents (PDFs) and outputting structured data (JSON) via APIs. * Basic operations of data cleaning and Pandas DataFrames. * Creating an automated workflow to build an exclusive literature review database for your Final Project.

2. Task Description

The teaching assistant has provided you with a basic sample code (see attachment/code block below). You need to modify this code so that it can read the specified PDF papers and extract the core dimensions.

Basic Requirements: The number of papers you ultimately process and successfully extract information from must be ≥ 10 . You can use the reference papers provided in the course, but it is strongly recommended that you replace them with literature relevant to your Final Project.

2.1 Output Schema Standard

Please explicitly ask the model in your Prompt to extract the following 8 fields (you can fine-tune them based on your topic). If the original text does not mention it explicitly, strictly instruct the model to output “N/A” or “None”.

No.	Field Name	Description & Example
1	Author & Year	Basic literature info (e.g., Strachan et al., 2024).
2	Core Research Topic	A one-sentence summary of the LLM capability or phenomenon studied in the paper.
3	Evaluated LLMs	Which specific models were tested in the research? (e.g., GPT-4, Claude 3, etc. If not specified, fill in “Not specified”).
4	Evaluation Tasks/Methods	What specific tasks were used for testing? (e.g., Turing test, economic games, role-playing prompt).
5	Human Baseline/Sample Size	Were human participants introduced for comparison? What was the human sample size?
6	Key Findings & Conclusions	What is the most important experimental conclusion of the paper?
7	AI Limitations or Risks	What issues with LLMs did the paper point out? (e.g., digital deception, moral risks of anthropomorphism, etc.).
8	Application Implications	What practical guidance does this research offer to the real world?

3. Deliverables & Submission Requirements

Please pack the following files into a single .zip archive and upload it to the assignment system, using the naming format Lab_StudentID_Name.zip:

1. Literature Summary Table (Core Output) (.csv or .xlsx): The DataFrame export file generated after the code runs successfully. It must contain complete extraction data for at least 10 papers.
2. Python Source Code (.py or .ipynb): Contains your fully modified code. (Please ensure you remove your real API Key before submitting!)
3. Short Lab Report (.pdf, max 2 pages):
 - Briefly describe the difficulties you encountered and your solutions when adjusting the Prompt or handling errors.
 - If you completed the bonus challenges, explain your implementation approach here.
 - Based on your generated literature summary table, write a brief paragraph summarizing the commonalities or core conclusions of the batch of literature you analyzed.